The Four Fundamental Subspaces

Every matrix carries four spaces: the targets you can reach, the trades that do nothing, everything you can measure, and the hidden link between your rules. One row reduction finds all four.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

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7. **Final memory table**
   All four side by side.

METHODS IN THIS LESSON: The four subspaces · Dimensions r, n − r, r, m − r · Row space basis from R · [A | I] → [R | E] · Left null space basis · Left null space test for b

PART 2 Prerequisites

Click any link to review it, then come back.

- Basis and dimension (Lesson 9): bases of $C(A)$ and $N(A)$, rank $= \dim C(A)$.
- R and special solutions (Lesson 7) and the zero-row test (Lesson 8).

- Transposes (Lesson 5) and Gauss–Jordan on [A | I] (Lesson 3).

PART 3 **Explanation**

THE FUNDS USED IN THE EXAMPLES

Columns (what ₹1 lakh does to budget, return, exposure): Nifty $(1, 12, 1)$, Nifty ETF $(1, 12, 1)$, bonds $(1, 7, 0)$, balanced $(1, 9.5, 0.5)$, midcap $(1, 15, 1.2)$. Rows are the rules: budget, return, exposure. So A is $m \times n$ with m = number of rules, n = number of funds.

1. Conditions and important concepts

1. The four subspaces

Any $m \times n$ matrix A gives four subspaces. Two you already know; two are new:

DEFINITION · THE FOUR FUNDAMENTAL SUBSPACES

Subspace	Made of	Lives in	In money terms
Column space $C(A)$	all combinations of the columns	$\mathbb{R}^m$	targets you can reach
Null space $N(A)$	all x with $Ax = 0$	$\mathbb{R}^n$	trades that do nothing
Row space $C(A^T)$	all combinations of the rows	$\mathbb{R}^n$	everything you can measure about a trade
Left null space $N(A^T)$	all y with $A^T y = 0$	$\mathbb{R}^m$	combinations of rules that are always zero: a hidden link

2. Column space C(A) (known)

All mixes of the funds' columns: the targets b for which $Ax = b$ can be solved. Basis: the pivot columns of A.

3. Null space N(A) (known)

All trades x with $Ax = 0$: they change no rule. Basis: the special solutions.

4. Row space C(Aᵀ) (new)

All combinations of the **rows** of A. A row is a rule; a combination of rows is a new rule, like "return $-12\times$ budget". So:

$$\boxed{\text{Row space} = \text{every measurement you can build from your rules.}}$$

The rows of A are the columns of A^T, so the row space is $C(A^T)$.

5. Left null space N(Aᵀ) (new)

All y with $A^T y = 0$. Taking the transpose of both sides, that is $y^T A = 0$: a row vector y^T on the **left** of A giving zero, hence the name. In words: a combination of the rules (weights y on the rows) that comes out to **zero for every fund**.

$$\boxed{\text{Left null space} = \text{combinations of rules that are always zero: a hidden link between the r}}$$

6. Where they live

$N(A)$ and $C(A^T)$ are made of vectors with n entries (one per fund): they live in $\mathbb{R}^n$. $C(A)$ and $N(A^T)$ are made of vectors with m entries (one per rule): they live in $\mathbb{R}^m$.

7. Dimensions

RULE · DIMENSIONS OF THE FOUR SUBSPACES

For an $m \times n$ matrix of rank r:

$$\dim C(A) = r \qquad \dim N(A) = n - r \qquad \dim C(A^T) = r \qquad \dim N(A^T) = m - r$$

8. Row rank = column rank

The row space and the column space have the **same dimension** r, even though they live in different places. Reason: every pivot sits in one pivot row and one pivot column, so both are counted by the same pivots.

9. Basis of the row space: the rows of R

METHOD · ROW SPACE BASIS = THE NON-ZERO ROWS OF R

Row operations replace rows by combinations of rows, and can be undone. So A and R have the **same row space**, and the first r (non-zero) rows of R are a basis.

IMPORTANT

For the **row** space you may use R's rows. For the **column** space you must use A's own pivot columns (Lesson 9). Row operations keep the row space but change the column space.

10. Finding the left null space: E

METHOD · REDUCE [A | I] TO [R | E]

Put the identity next to A and row reduce the whole thing. The right block becomes the matrix E that records every row operation, so $EA = R$. (If A is square and invertible, $R = I$ and $E = A^{-1}$, as in Gauss–Jordan.)

METHOD · LEFT NULL SPACE BASIS = THE ROWS OF E NEXT TO THE ZERO ROWS OF R

If row i of R is all zeros, then (row i of E) $\times A$ = row i of R = 0. So the last $m - r$ rows of E are a basis of $N(A^T)$.

11. What the left null space tells you: which targets are reachable

If $y^T A = 0$, then for any portfolio x: $y^T(Ax) = (y^T A)x = 0$. So every reachable target $b = Ax$ must have $y^T b = 0$. And that is also enough:

RULE · THE LEFT NULL SPACE TEST

$Ax = b$ is solvable $\iff$ $y^T b = 0$ for every y in $N(A^T)$. This is Lesson 8's zero-row test, written as a vector.

12. The two pairs

The dimensions add up in pairs:

$$\boxed{\dim C(A^T) + \dim N(A) = r + (n - r) = n} \qquad \boxed{\dim C(A) + \dim N(A^T) = r + (m - r)}$$

13. Square and invertible

If A is $n \times n$ with rank n: $C(A) = C(A^T) = \mathbb{R}^n$, and $N(A) = N(A^T) = \{0\}$. No do-nothing trade, no hidden link, every target reachable in one way.

2. Procedure: all four in one go

Step	What to do	Gives
1	Write $[A \mid I]$ (I is $m \times m$).	
2	Row reduce to $[R \mid E]$.	R, E with $EA = R$
3	Count the pivots.	rank r
4	Pivot columns of A.	basis of $C(A)$, dim r
5	Special solutions from R.	basis of $N(A)$, dim $n - r$
6	Non-zero rows of R.	basis of $C(A^T)$, dim r
7	Rows of E next to the zero rows of R.	basis of $N(A^T)$, dim $m - r$
8	Check $r + (n - r) = n$ and $r + (m - r) = m$.	

3. 2 × 2 example

Nifty and its twin ETF, with the budget and return rules:

$$[\,A \mid I\,] = \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 12 & 12 & 0 & 1 \end{array}\right] \xrightarrow{\text{row } 2 - 12 \times \text{row } 1} \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 0 & -12 & 1 \end{array}\right] = [\,R \mid E\,]$$

(Row 2: $12 - 12 = 0, 12 - 12 = 0, 0 - 12(1) = -12, 1 - 12(0) = 1$.) One pivot: $r = 1, m = n = 2$.

Subspace	Basis	Dim	Meaning
$C(A)$	$(1, 12)$ (column 1 of A)	1	only targets with return $= 12\times$ budget
$N(A)$	$(-1, 1)$	1	swap Nifty for its twin
$C(A^T)$	$(1, 1)$ (row 1 of R)	1	the only real measurement: total Nifty held, $x_1 + x_2$
$N(A^T)$	$(-12, 1)$ (row 2 of E)	1	the hidden link: $-12 \cdot \text{budget} + \text{return} = 0$

Check the left null vector: $y^T A = (-12, 1)\begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix} = (-12 + 12,\ -12 + 12) = (0, 0)$ ✓. So every reachable target has return $= 12\times$ budget: $(10, 120)$ passes ($-120 + 120 = 0$), $(10, 105)$ fails ($-120 + 105 = -15$), exactly Lesson 8's answer.

Check the pairs: $1 + 1 = 2 = n$ ✓, $1 + 1 = 2 = m$ ✓.

4. 3 × 3 example

Nifty, bonds, balanced, all three rules:

$$[\,A \mid I\,] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 12 & 7 & 9.5 & 0 & 1 & 0 \\ 1 & 0 & 0.5 & 0 & 0 & 1 \end{array}\right]$$

The same five row operations as Lesson 7. The A part changes exactly as before; here is what happens to the identity part:

Row operation	A part (as before)	Identity part becomes
Row 2 $-$ 12$\times$ row 1	$(0, -5, -2.5)$	row 2: $(0-12, 1-0, 0-0) = (-12, 1, 0)$
Row 3 $-$ 1$\times$ row 1	$(0, -1, -0.5)$	row 3: $(0-1, 0, 1) = (-1, 0, 1)$
Row 3 $-$ 0.2$\times$ row 2	$(0, 0, 0)$	row 3: $(-1+2.4,\ 0-0.2,\ 1) = (1.4, -0.2, 1)$
Row 2 $\div(-5)$	$(0, 1, 0.5)$	row 2: $(-12/-5,\ 1/-5,\ 0) = (2.4, -0.2, 0)$
Row 1 $-$ 1$\times$ row 2	$(1, 0, 0.5)$	row 1: $(1-2.4,\ 0+0.2,\ 0) = (-1.4, 0.2, 0)$

$$[R \mid E] = \left[\begin{array}{ccc|ccc} \boxed{1} & 0 & 0.5 & -1.4 & 0.2 & 0 \\ 0 & \boxed{1} & 0.5 & 2.4 & -0.2 & 0 \\ 0 & 0 & 0 & 1.4 & -0.2 & 1 \end{array}\right] \qquad r = 2,\ m = n = 3$$

Subspace	Basis	Dim	Meaning
$C(A)$	Nifty $(1, 12, 1)$, bonds $(1, 7, 0)$	2	the plane of reachable targets
$N(A)$	$(-0.5, -0.5, 1)$	1	build balanced from Nifty and bonds
$C(A^T)$	$(1, 0, 0.5)$, $(0, 1, 0.5)$ (rows of R)	2	"Nifty content" $x_1 + 0.5x_3$ and "bond content" $x_2 + 0.5x_3$
$N(A^T)$	$(1.4, -0.2, 1)$, or $\times 5$: $(7, -1, 5)$	1	the hidden link $7B - R + 5E = 0$

Reading the row space

The two rows of R are two measurements of a trade: how much Nifty it really holds (Nifty plus half the balanced) and how much bonds it really holds. Every rule is a combination of them:

- Budget $(1, 1, 1) = 1 \cdot (1, 0, 0.5) + 1 \cdot (0, 1, 0.5)$ ✓ $(0.5 + 0.5 = 1)$.
- Return $(12, 7, 9.5) = 12 \cdot (1, 0, 0.5) + 7 \cdot (0, 1, 0.5)$ ✓ $(6 + 3.5 = 9.5)$.
- Exposure $(1, 0, 0.5) = 1 \cdot (1, 0, 0.5)$ ✓.

Three rules, but only **two** independent measurements: that's $\dim C(A^T) = 2$.

Reading the left null space

Check $y = (7, -1, 5)$ against every fund (each column of A):

- Nifty: $7(1) - 1(12) + 5(1) = 7 - 12 + 5 = 0$ ✓
- Bonds: $7(1) - 1(7) + 5(0) = 0$ ✓
- Balanced: $7(1) - 1(9.5) + 5(0.5) = 7 - 9.5 + 2.5 = 0$ ✓

So for **every** portfolio, $7 \cdot \text{budget} - \text{return} + 5 \cdot \text{exposure} = 0$, which is $R = 7B + 5E$. The left null space found the hidden link between the rules, and with it the test for targets:

- $(10, 105, 7)$: $70 - 105 + 35 = 0$ ✓ reachable.
- $(10, 105, 5)$: $70 - 105 + 25 = -10$ ✗ not reachable.

Check the pairs: $2 + 1 = 3 = n$ ✓, $2 + 1 = 3 = m$ ✓.

▼ For the curious: and with midcap instead of balanced?

Nifty, bonds, midcap is invertible ($r = 3$), so $R = I$ and $E = A^{-1}$. $C(A) = C(A^T) = \mathbb{R}^3$ and $N(A) = N(A^T) = \{0\}$: no do-nothing trade, no hidden link, every target reachable exactly once.

5. Bigger example: four funds

Nifty, ETF, bonds, balanced: $m = 3$ rules, $n = 4$ funds. Row reduction uses the same five operations, so E comes out the same as in the 3 × 3 example:

$$[\,R \mid E\,] = \left[\begin{array}{cccc|ccc} \boxed{1} & 1 & 0 & 0.5 & -1.4 & 0.2 & 0 \\ 0 & 0 & \boxed{1} & 0.5 & 2.4 & -0.2 & 0 \\ 0 & 0 & 0 & 0 & 1.4 & -0.2 & 1 \end{array}\right] \qquad r = 2$$

Now **all four** subspaces are non-trivial:

Subspace	Lives in	Basis	Dim
$C(A)$	$\mathbb{R}^3$	Nifty $(1, 12, 1)$, bonds $(1, 7, 0)$	$r = 2$
$N(A)$	$\mathbb{R}^4$	$(-1, 1, 0, 0)$, $(-0.5, 0, -0.5, 1)$	$n - r = 2$
$C(A^T)$	$\mathbb{R}^4$	$(1, 1, 0, 0.5)$, $(0, 0, 1, 0.5)$	$r = 2$
$N(A^T)$	$\mathbb{R}^3$	$(1.4, -0.2, 1)$	$m - r = 1$

Meaning. The row space now measures "Nifty content" $x_1 + x_2 + 0.5x_4$ (Nifty, plus the ETF, plus half the balanced) and "bond content" $x_3 + 0.5x_4$. The do-nothing trades are exactly the trades that leave both measurements at 0: check $s_1 = (-1, 1, 0, 0)$: $-1 + 1 + 0 = 0$ and $0 + 0 = 0$ ✓; $s_2 = (-0.5, 0, -0.5, 1)$: $-0.5 + 0 + 0.5 = 0$ and $-0.5 + 0.5 = 0$ ✓. The left null vector checks out for the ETF too: $7(1) - 12 + 5(1) = 0$ ✓.

Check the pairs: $2 + 2 = 4 = n$ ✓, $2 + 1 = 3 = m$ ✓.

Try it: pick a menu and see all four

☑ Nifty ☑ ETF ☑ Bonds ☑ Balanced ☐ Midcap

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 \\ 1 & 1 & 0 & 0.5 \end{bmatrix}$$

$[R \mid E]$: $R = \begin{bmatrix} 1 & 1 & 0 & 0.5 \\ 0 & 0 & 1 & 0.5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $E = \begin{bmatrix} -1.4 & 0.2 & 0 \\ 2.4 & -0.2 & 0 \\ 1.4 & -0.2 & 1 \end{bmatrix}$. Rank $r = 2$ ($m = 3$ rules, $n = 4$ columns).

Subspace	Basis	Dim
$C(A)$ in $\mathbb{R}^3$	Nifty $(1, 12, 1)$, Bonds $(1, 7, 0)$	2
$N(A)$ in $\mathbb{R}^4$	$(-1, 1, 0, 0)$, $(-0.5, 0, -0.5, 1)$	2
$C(A^T)$ in $\mathbb{R}^4$	$(1, 1, 0, 0.5)$, $(0, 0, 1, 0.5)$	2
$N(A^T)$ in $\mathbb{R}^3$	$(7, -1, 5)$	1

Hidden link: 7 budget − 1 return + 5 exposure = 0 for every portfolio. Check against each column: 0, 0, 0, 0 ✓

Pairs: $2 + 2 = 4$ and $2 + 1 = 3$ ✓

6. Connection: the big picture

$$\underbrace{\mathbb{R}^n}_{\text{trades}} : \underbrace{C(A^T)}_{\dim r \text{ (measured)}} + \underbrace{N(A)}_{\dim n-r \text{ (invisible)}} \qquad \underbrace{\mathbb{R}^m}_{\text{targets}} : \underbrace{C(A)}_{\dim r \text{ (reachable)}} + \underbrace{N(A^T)}_{\dim m-r \text{ (hidden links)}}$$

On the **trade side** ($\mathbb{R}^n$): each direction is either something your rules can see (row space) or something they can't (null space). On the **target side** ($\mathbb{R}^m$): each direction is either reachable

(column space) or blocked by a hidden link (left null space).

Menu	m	n	r	$\dim N(A)$	$\dim N(A^T)$	So
Nifty, twin (2 rules)	2	2	1	1	1	return locked to 12 × budget
Nifty, bonds, balanced	3	3	2	1	1	$R = 7B + 5E$ locked
Nifty, ETF, bonds, balanced	3	4	2	2	1	$R = 7B + 5E$ locked
All five (+ midcap)	3	5	3	2	0	no hidden link: every target reachable
Nifty, bonds, midcap	3	3	3	0	0	every target, exactly one way

READING IT AS A TRADER

The left null space is the market's **hidden constraint**: with only Nifty-and-bond-like products, return and exposure can't be chosen separately. Adding a product with a genuinely new direction (midcap) kills the left null space and frees every target. A market where $N(A^T) = \{0\}$, i.e. the products' payoffs span every scenario, is what finance calls a **complete market**.

7. Final memory table

	Column space $C(A)$	**Null space** $N(A)$	**Row space** $C(A^T)$	**Left null space** $N(A^T)$
Made of	mixes of columns	x with $Ax = 0$	mixes of rows	y with $y^T A = 0$
Lives in	$\mathbb{R}^m$	$\mathbb{R}^n$	$\mathbb{R}^n$	$\mathbb{R}^m$
Dimension	r	$n - r$	r	$m - r$
Basis	pivot columns of A	special solutions	non-zero rows of R	rows of E at R's zero rows
Money meaning	reachable targets	do-nothing trades	what the rules can measure	hidden links between rules
Example (4 funds)	Nifty, bonds	s_1, s_2	Nifty content, bond content	$(7, -1, 5)$

Pairs: $r + (n - r) = n$ and $r + (m - r) = m$. Find everything with one reduction: $[A \mid I] \to [R \mid E]$.

▼ **Check yourself (MIT problem 10.1): A is $m \times n$ of rank r, and some $Ax = b$ has no solution. Which inequalities must hold? Why does $A^T y = 0$ have non-zero solutions?**

$r \leq m$ and $r \leq n$ always. "Some b unreachable" means $C(A) \neq \mathbb{R}^m$, so $r < m$. Then $\dim N(A^T) = m - r > 0$, so the left null space has non-zero vectors: the hidden link that blocks those targets.

▼ **Check yourself (MIT lecture): the four subspaces of $\begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$.**

$R = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, $E = \begin{bmatrix} -1 & 2 & 0 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$, $r = 2$. $C(A)$: columns 1, 2 of A (dim 2). $N(A)$: $(-1, -1, 1, 0)$, $(-1, 0, 0, 1)$ (dim 2). $C(A^T)$: $(1, 0, 1, 1)$, $(0, 1, 1, 0)$ (dim 2). $N(A^T)$: $(-1, 0, 1)$ (dim 1): row 3 minus row 1 is zero, since rows 1 and 3 are equal.